

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2019

Subject: 353152 Electronic Circuits I

Section: 05-06

Date: 17 October 2019

Time: 13.00-15.00

Name: _____ ID: _____ Field of Study: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 11 pages (including this page) and they are worth 80 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

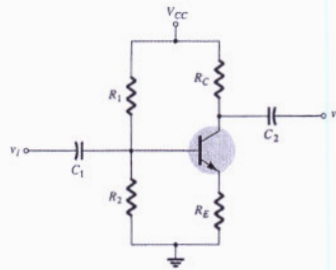
Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

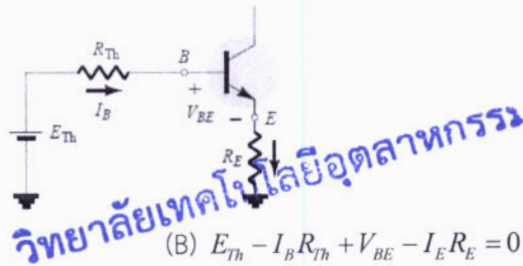
Part 1 (50 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

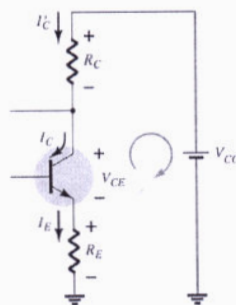
1. What is the DC-bias type of the circuit in Figure below?



- (A) Fixed bias
(B) Emitter-stabilized bias
(C) Voltage divider
(D) Voltage feedback
2. What is the KVL equation of the base-emitter loop of Figure below?

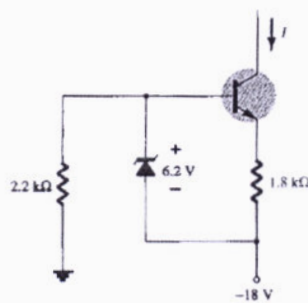


- (A) $E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$
(B) $E_{Th} - I_B R_{Th} + V_{BE} - I_E R_E = 0$
(C) $E_{Th} + I_B R_{Th} + V_{BE} - I_E R_E = 0$
(D) $-E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$
3. From Collector-Emitter loop below, find the voltage equation of this closed loop.



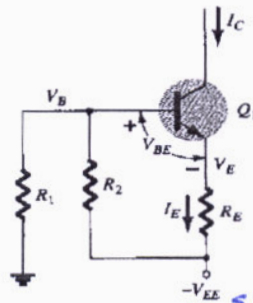
- (A) $I_E R_E + V_{CE} + I_E R_C = V_{CC}$
(B) $I_E R_E - V_{CE} - I_E R_C = V_{CC}$
(C) $I_E R_E - V_{CE} + I_E R_C = V_{CC}$
(D) $I_E R_E - V_{CE} + I_E R_C = -V_{CC}$

4. Find constant current (A) of the transistor/Zener Constant-Current Source in Figure below.



- (A) 3.06 mA (B) 1.06 mA (C) 2.06 mA (D) 5.06 mA

5. What is the V_B voltage equation in Figure below?



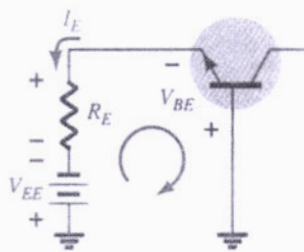
(A) $V_B = \frac{R_2}{R_1 + R_2} (-V_{EE})$

(B) $V_B = \frac{R_1}{R_2} (-V_{EE})$

(C) $V_B = \frac{R_1}{2R_2} (-V_{EE})$

(D) $V_B = \frac{R_1}{R_1 + R_2} (-V_{EE})$

6. From base-Emitter loop below, find the KVL voltage equation of this closed loop.



(A) $V_{EE} + I_E R_E = V_{BE}$

(B) $V_{EE} - I_E R_E = -V_{BE}$

(C) $-V_{EE} + I_E R_E = V_{BE}$

(D) $V_{EE} - I_E R_E = V_{BE}$

7. In Collector-Feedback configuration, we assume that the $I'_C \cong I_C$, $I_E \cong I_C$. What is the reason of the assumption?

(A) Because of $I_B \ll I_C$

(B) Because of $I_B = I_C$

(C) Because of $I_B \gg I_C$

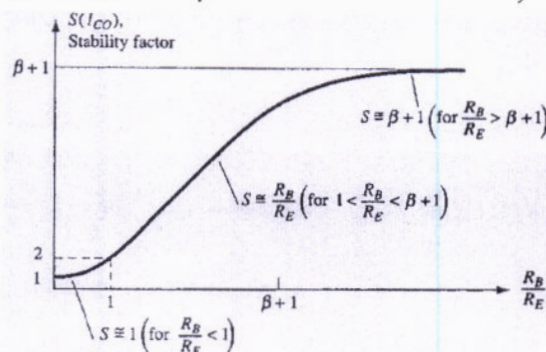
(D) Because of $I_B \approx I_C$

8. In a fixed-bias circuit, the slope of the dc load line is controlled by ____.

- (A) R_B (B) R_C (C) V_{CC} (D) V_{BE}

9. The stability factor $S(I_{CO})$ of Emitter-Bias configuration is shown in Figure below.

Given $R_B = 500 \text{ k}\Omega$, $R_E = 5 \text{ k}\Omega$ and $\beta = 100$. What is the stability factor $S(I_{CO})$ value?



- (A) 101 (B) 100 (C) 501 (D) 500

10. Which class of power amplifiers that conducts less than 360° of input?

- (A) class A (B) class B (C) class AB (D) class C

11. Which class of power amplifiers having Q-point at the middle of active region?

- (A) class A (B) class B (C) class AB (D) class C

12. How can the class A amplifier efficiency be improved from 25%?

- (A) increase input current. (B) reduce input current.
(C) add transformer at input side. (D) add transformer at output side.

13. Which class of power amplifiers having power efficiency up to 78.5%

- (A) class A (B) class B (C) class AB (D) class C

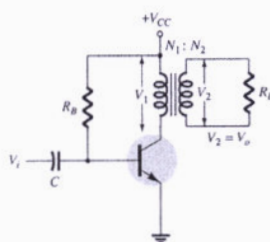
14. Which class of power amplifiers is used for the digital logic gate.

- (A) class A (B) class D (C) class AB (D) class C

15. Which type of following power amplifiers having the highest efficiency?

- (A) class A (B) class B (C) class AB (D) class C

16. What the propose of the transformer-coupled class A amplifier?



- (A) Improve power efficiency (B) Reduce noise
(C) Reject undesired frequency (D) Increase stability factor

17. What is the value of $P_o(AC)$ of class B amplifier if $V_L(p-p) = 10\text{ V}$ and $R_L = 2\text{ ohms}$.
 (A) 50 W (B) 25 W (C) 6.25 W (D) 5.25 W
18. What is the purpose of using push-pull circuit for class B amplifier?
 (A) to obtain full cycle (B) to reduce the distortion
 (C) to increase speed of signal (D) to reduce the cost of amplifier
19. Which of the following is (are) the application(s) of a transistor?
 (A) Signal protector (B) Voltage regulator
 (C) Computer logic circuitry (D) DC Adapter
20. In an emitter-bias configuration, the _____ the resistance R_E , the _____ the stability factor, and the _____ stable is the system.
 (A) smaller, lower, less (B) larger, more, more
 (C) smaller, more, more (D) larger, lower, more
21. The _____ the stability factor, the _____ sensitive the network is to variations in that parameter.
 (A) higher, more (B) higher, less
 (C) lower, more (D) None of the above
22. In a transistor-switching network, the level of the resistance between the collector and emitter is _____ at the saturation and is _____ at the cutoff.
 (A) low, low (B) low, high (C) high, high (D) high, low
23. _____ is the least stabilized circuit.
 (A) Fixed bias (B) Emitter-stabilized bias
 (C) Voltage divider (D) Voltage feedback
24. Changes in temperature will affect the level of _____.
 (A) current gain β (B) leakage current I_{CEO}
 (C) both current gain β and leakage current I_{CEO} (D) None of the above

25. In a voltage-divider circuit, which one of the stability factors has the least effect on the device at very high temperature?

(A) $S(I_{CO})$

(B) $S(V_{BE})$

(C) $S(\beta)$

(D) Undefined

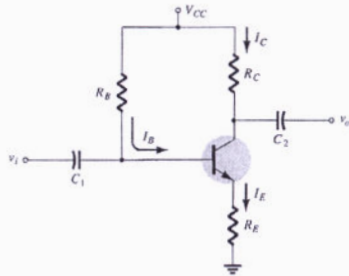
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Part 2 (30 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

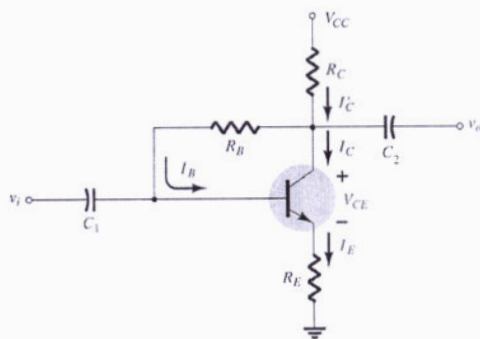
1. (5 point) From the Emitter-bias configuration below, given $R_B=430\text{k}\Omega$, $R_C=2\text{k}\Omega$, $R_E=1\text{k}\Omega$ and $\beta=50$.

Find the value of (a) I_B and I_C (b) V_{CE} , V_C , V_E and V_B



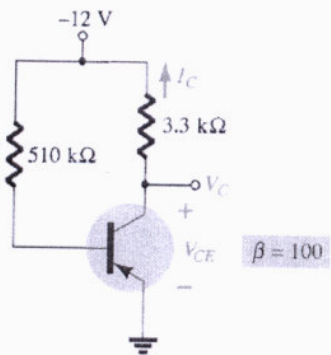
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2. (5 points) Given $R_C=5\text{k}\Omega$, $R_B=100\text{k}\Omega$, $R_E=10\text{k}\Omega$ and $V_{CC}=18\text{V}$, find the current I_B and voltage V_{CE} .



$$\beta=120$$

3. (5 points) Determine the V_C , V_{CE} and I_C



4. (5 points) From the table below, determine the current ΔI_C if the temperature is changed from 25 °C to 175°C. Given $R_B = 500 \text{ k}\Omega$ and $R_E = 50 \text{ k}\Omega$.

Given:

$$S(I_{CO}) = (\beta + 1) \frac{1 + R_B/R_E}{(\beta + 1) + R_B/R_E}, \text{ and } S(V_{BE}) = \frac{-\beta}{R_B + (\beta + 1)R_E}$$
$$\Delta I_C = S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE}$$

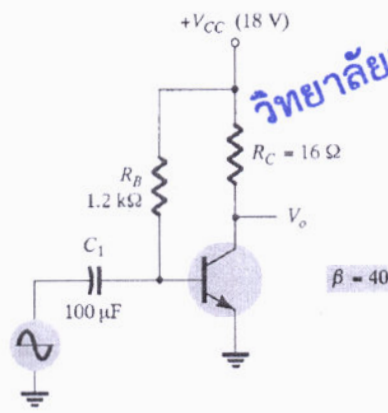
$T \text{ (}^\circ\text{C)}$	$I_{CO} \text{ (nA)}$	β	$V_{BE} \text{ (V)}$
-65	0.2×10^{-3}	20	0.85
25	0.1	50	0.65
100	20	80	0.48
175	3.3×10^3	120	0.3

5. (5 points) For a class B amplifier, determine the efficiency if a supply voltage of $V_{CC}=22\text{ V}$ driving a $4\ \Omega$ load with peak output voltages of

6.1 $V_{L(p)}=20\text{ V}$

6.2 $V_{L(p)}=4\text{ V}$

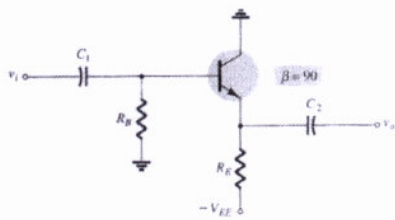
6. (5 points) Calculate the input and output power for the circuit below, if the input signal results in a base current of 5 mA(rms) .



Instructors: Assoc. Prof. Dr. Titipong Lertwiriaprapa
and Asst. Prof. Dr. Kittisak Phaebua

Handout

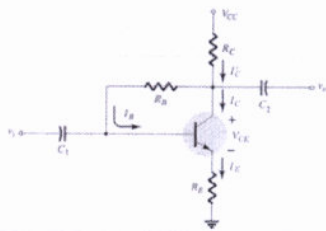
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

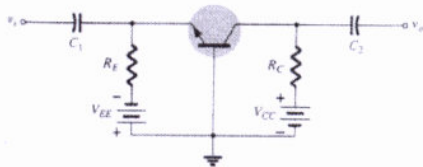
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_o(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\% \eta = 50 \left(\frac{V_{CE(max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(min)}} \right)^2$$

$$\frac{R'_L}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2} \right)^2 = a^2$$

Class B amplifier

$$P_o(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC} \left(\frac{2}{\pi} I_L(p) \right)$$

$$\% \eta = \frac{P_o(ac)}{P_i(dc)} \times 100\%$$

Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

	(A)	(B)	(C)	(D)
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